

Periodic Hunter–Saxton Flow: Exact Two-Sided Lifespan and Universal Wave Breaking

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Abstract

For every nonconstant periodic C^2 datum, we solve the once-integrated Hunter–Saxton equation throughout its maximal classical characteristic interval. One scalar factor gives the exact Jacobian, both lifespan endpoints, and every first-breaking label. We prove conservation of slope energy and the universal coefficient -2 at all simultaneous first breaking points, and close the constant-data and post-breaking boundaries.

1 Exact characteristic theorem

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and consider the once-integrated equation

$$u_{tx} + uu_{xx} + \frac{1}{2}u_x^2 = -\frac{1}{2}E, \quad E = \int_{\mathbb{T}} u_x(t, x)^2 dx. \quad (1)$$

This is our classical formulation for C^2 data; the differentiated third-order equation is not asserted pointwise at that regularity. Put $w_0 = u_{0x}$, $m = \min w_0$, and $M = \max w_0$. Nonconstant periodicity implies $E > 0$ and $m < 0 < M$.

Theorem 1 (Complete pre-breaking dynamics). *Define, for a characteristic label a ,*

$$F(t, a) = \cos \frac{\sqrt{E}t}{2} + \frac{w_0(a)}{\sqrt{E}} \sin \frac{\sqrt{E}t}{2}, \quad (2)$$

$$T_+ = \frac{2}{\sqrt{E}} \arctan \frac{\sqrt{E}}{-m}, \quad T_- = -\frac{2}{\sqrt{E}} \arctan \frac{\sqrt{E}}{M}. \quad (3)$$

There is a degree-one characteristic diffeomorphism $\eta(t, \cdot)$ for exactly $T_- < t < T_+$, normalized up to a common translation, such that

$$\eta_a = F^2, \quad u_x(t, \eta(t, a)) = \frac{2F_t}{F} = \frac{-\sqrt{E} \sin(\sqrt{E}t/2) + w_0(a) \cos(\sqrt{E}t/2)}{F(t, a)}. \quad (4)$$

The first positive breaking labels are exactly $\arg\min w_0$; all of them break simultaneously. At each such label,

$$u_x(t, \eta(t, a)) = -\frac{2}{T_+ - t} + O(1) \quad (t \uparrow T_+). \quad (5)$$

Backward degeneration occurs exactly at $\arg\max w_0$ and time T_- . No earlier degeneration occurs in either direction.

Proof. Equation (1) yields along $\eta_t = u(t, \eta)$ the Riccati equation

$$\dot{w} = -\frac{1}{2}(w^2 + E), \quad w(t, a) = u_x(t, \eta(t, a)).$$

Its solution is $w = 2F_t/F$. Since $\dot{\eta}_a = w\eta_a$ and $\eta_a(0, a) = 1$, this gives $\eta_a = F^2$. Moreover

$$\int_{\mathbb{T}} F^2 da = \cos^2 \frac{\sqrt{E}t}{2} + \sin^2 \frac{\sqrt{E}t}{2} = 1,$$

because $\int w_0 = 0$ and $\int w_0^2 = E$. Conversely this identity makes the construction explicit: set

$$\eta(t, a) = b(t) + \int_0^a F(t, s)^2 ds, \quad u(t, \eta(t, a)) = \eta_t(t, a),$$

where $b(0) = 0$ and $b'(0) = u_0(0)$. Then $\eta(0, a) = a$, $u(0, a) = u_0(a)$, and differentiating in a gives $u_x \circ \eta = \eta_{ta}/\eta_a = 2F_t/F$. Its Riccati equation is precisely (1) along every characteristic. The free common translation $b(t)$ is the harmless additive gauge. Thus the construction is a degree-one circle diffeomorphism exactly while F is positive.

For fixed $t > 0$ before the first pole, F is increasing in w_0 . Its first zero is therefore attained precisely where $w_0 = m$, and solving $F = 0$ gives T_+ . Reversing time makes the maximum decisive and gives T_- . At a future first label, $F(T_+, a) = 0$ and $F_t(T_+, a) \neq 0$. Thus $2F_t/F = 2/(t - T_+) + O(1)$, proving (5) and the complete label statement. $\square$

2 Energy, degeneracies, and the endpoint

The same representation closes the conservation law without dividing by the Jacobian at its endpoint:

$$w(t, a)^2 \eta_a(t, a) = E \left[-\sin \frac{\sqrt{E}t}{2} + \frac{w_0(a)}{\sqrt{E}} \cos \frac{\sqrt{E}t}{2} \right]^2. \quad (6)$$

Integrating (6), the mixed term vanishes and the remaining two terms sum to E . Consequently $\int_{\mathbb{T}} u_x^2 dx = E$ throughout (T_-, T_+) , consistently closing (1).

If $E = 0$, then $w_0 = 0$ and the solution is spatially constant; neither formula in (3) is interpreted by division through zero. A flat interval or several isolated points may realize the global minimum, and Theorem 1 includes the entire minimizing set rather than a generic singleton. At $t = T_+$ the map ceases to be a diffeomorphism. This paper neither constructs nor selects a conservative, dissipative, or other weak continuation.

Source lineage

References

- [1] J. K. Hunter and R. Saxton, “Dynamics of director fields,” *SIAM J. Appl. Math.* 51 (1991), 1498–1521. DOI: [10.1137/0151075](https://doi.org/10.1137/0151075).
- [2] J. Lenells, “The Hunter–Saxton Equation: A Geometric Approach,” *SIAM J. Math. Anal.* 40 (2008), 266–277. DOI: [10.1137/050647451](https://doi.org/10.1137/050647451).
- [3] Z. Yin, “On the Structure of Solutions to the Periodic Hunter–Saxton Equation,” *SIAM J. Math. Anal.* 36 (2004), 272–283. DOI: [10.1137/S0036141003425672](https://doi.org/10.1137/S0036141003425672).